



Programming Language

Four criteria

- readability
- writability
- reliability
- cost

Hybrid implementation system

Fortran - scientific computing

Algol 60 - first syntax

Backus - Naur form

Parse tree - ambiguous
pairwise disjointness test
reserved word

SOLV

- Consider the below grammar. Show that the string **bbacaabb** write leftmost derivation

$$\langle S \rangle \Rightarrow \langle A \rangle c \langle B \rangle b$$

$$\langle A \rangle \Rightarrow b \langle A \rangle | a$$

$$\langle B \rangle \Rightarrow a \langle B \rangle | b$$

Given

$$\langle S \rangle \Rightarrow \langle A \rangle c \langle B \rangle b$$

$$\Rightarrow b \langle A \rangle c \langle B \rangle b$$

$$\Rightarrow bb \langle A \rangle c \langle B \rangle b$$

$$\Rightarrow bba c \langle B \rangle b$$

$$\Rightarrow bba ca \langle B \rangle b$$

$$\Rightarrow bba caa \langle B \rangle b$$

$$\Rightarrow bba caa bb$$

Solu $\Rightarrow$ Re-write the below grammar rule by eliminating direct left recursive so that it is possible to develop a recursive descent parser for the same language.

$$\begin{aligned}\langle A \rangle &\rightarrow \langle A \rangle * \langle B \rangle \mid \langle A \rangle / \langle B \rangle \mid a \\ \langle B \rangle &\rightarrow \langle B \rangle + \langle C \rangle \mid \langle B \rangle - \langle C \rangle \mid b \\ \langle C \rangle &\rightarrow c\end{aligned}$$

Cözüm $\Rightarrow$

$$\begin{aligned}\langle A \rangle &\rightarrow a \langle A' \rangle \\ \langle A' \rangle &\rightarrow * \langle B \rangle \langle A' \rangle \mid / \langle B \rangle \langle A' \rangle \mid \epsilon \\ \langle B \rangle &\rightarrow b \langle B' \rangle \\ \langle B' \rangle &\rightarrow + \langle C \rangle \langle B' \rangle \mid - \langle C \rangle \langle B' \rangle \mid \epsilon \\ \langle C \rangle &\rightarrow c\end{aligned}$$

Solu

First function called (y)
then y called function (question)
question called (f)

```
int b;  
void question () {  
    int i, k  
    ①  
    void f () {  
        int b, k, m;  
        ② 3 3  
    }  
}  
void y () {  
    int m, j 3  
    ③
```

Coding

Static scope

Refers enviro

Line 1 $\Rightarrow$ Global b.
i, k of question

Line 2 $\Rightarrow$ i of question
b, k, m of f

Line 3 $\Rightarrow$ Global b
m, j of y

Static scope $\Rightarrow$ fonksiyon çağırma sırasının
özeni yoktur.

Dynamic scope $\longrightarrow$ first (y)
question
(f)

1 $\Rightarrow$ Global b,
m, j of y
i, k of question

2 $\Rightarrow$ j of y
i of question
b, k, m of y

3 $\Rightarrow$ Global b
m, j of y

Sol 1)

```
int i, j;
void { int k, *A, *B;
    static int x;
    A = malloc (sizeof) * 100;
    B = malloc (sizeof) * 200;
    A = B;
    free(B);
}
```

Ques

Life time

P = static

J = static

k = stack dynamic

A = Explicit heap dynamic

B = Explicit heap dynamic

x = static.

Two problem

A = B // A and B

becomes alias and

the heap dynamic
variable pointed by

Store 100 integer

becomes lost heap
dynamic variable

free(B)

A become dangling

pointer, because it

points to

a heap dynamic variable
that is deallocated.